

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	ARAVIND KUMAR.S
Register Number	192524210

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

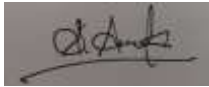
Total Marks: 50

Code of Conduct

I **ARAVIND KUMAR.S/192524210** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

ANSWER:

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet, allowing users to access resources on demand without owning physical infrastructure.

Characteristics:

- **On-Demand Self-Service:** Users can provision computing resources whenever needed without human intervention.
- **Broad Network Access:** Services are available over the internet and can be accessed from various devices.
- **Resource Pooling:** Computing resources are shared among multiple users using a multi-tenant model.
- **Rapid Elasticity:** Resources can be scaled up or down quickly based on workload requirements.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing evolved through several stages:

- **Hardware Evolution:** Mainframes and personal computers introduced computing capabilities.
- **Networking:** Local Area Networks (LANs) and the Internet enabled communication between systems.
- **Distributed Computing:** Resources were shared across multiple computers for better performance.
- **Virtualization:** Physical servers were divided into multiple virtual machines, improving resource utilization.
- **Internet Software Evolution:** Web applications, Service-Oriented Architecture (SOA), and Web Services enabled software delivery over the internet.
- **Cloud Platforms:** These technologies combined to create scalable, on-demand cloud services such as AWS, Microsoft Azure, and Google Cloud.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization:

Server virtualization allows multiple virtual machines (VMs) to run on a single physical server. It improves hardware utilization, reduces costs, simplifies management, and enables efficient resource allocation.

Difference:

Virtualization	Cloud Computing
An enabling technology that creates virtual resources.	A service model that delivers computing resources over the internet.
Focuses on efficient use of physical hardware.	Focuses on providing on-demand services to users.
Can exist without cloud computing.	Usually relies on virtualization to provide scalable services.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Feature	SOAP	REST
Communication Style	Protocol-based	Architectural style
Data Exchange	XML only	JSON, XML, HTML, and others
Performance	Slower due to XML processing	Faster and lightweight
Flexibility	More rigid with strict standards	Simple and flexible
Cloud Suitability	Suitable for enterprise applications requiring high security	Ideal for cloud and web applications due to speed and simplicity

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Service Model	Description	Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage, and networking infrastructure.	Amazon EC2
PaaS (Platform as a Service)	Provides a platform to develop, test, and deploy applications.	Google App Engine
SaaS (Software as a Service)	Delivers ready-to-use software over the internet.	Microsoft 365
XaaS (Anything as a Service)	Offers various IT services through the cloud, such as storage, security, or databases.	Dropbox (Storage as a Service)
